

Explanation of the Combined Pipeline & Results Figure

1. Purpose of This Note

This document explains, panel by panel, the combined figure prepared for the “Result Discussion” subsection of the paper. The goal of the figure is to give a reader the entire pipeline from raw Raman acquisition through to final classification performance in a single, self-contained visual, so that the surrounding text can focus on interpretation rather than re-describing the method from scratch. The sections below walk through what each panel shows, how it was generated, and why it was included.

2. The Figure

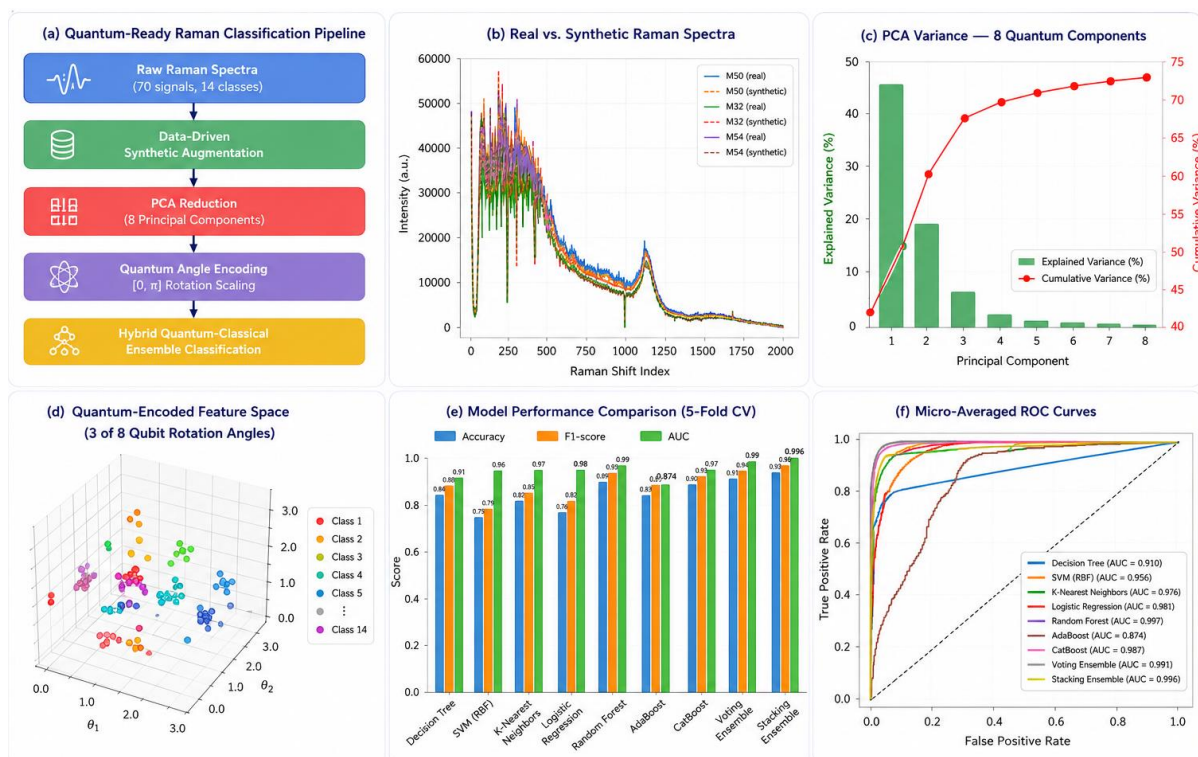


Figure 2. Overview of the proposed quantum-inspired Raman glucose-classification framework and its performance.

3. What the Overall Pipeline Is Doing

The dataset consists of only 70 real Raman spectra spread across 14 glucose concentration classes (5 spectra per class), which is far too small to train reliable classifiers on its own. To address this, the framework goes through five stages before any classifier ever sees the data:

- **Raw acquisition** we start with the raw, real spectra as acquired;
- **Synthetic augmentation** a class-specific statistical model (log-domain mean spectrum + scaled residual + feature-wise noise) is used to generate 1,000 synthetic spectra per class, expanding 70 real samples into 14,000 samples while preserving each class's true spectral shape;
- **PCA reduction** the augmented spectra are enhanced (gradient-based sharpening of glucose-sensitive peaks) and reduced from 2,030 raw intensity values down to 8 principal components, chosen because we plan to eventually encode these onto 8 qubits;

- **Quantum angle encoding** each of the 8 PCA components is rescaled into the $[0, \pi]$ range so it can be directly used as a rotation angle for a quantum gate (R_x , R_y , or R_z) — this is the step that makes the representation “qubit-ready”;
- **Classification** nine classifiers (Decision Tree, SVM, KNN, Logistic Regression, Random Forest, AdaBoost, CatBoost, a Voting Ensemble, and a Stacking Ensemble) are trained and compared on this 8-dimensional representation using stratified 5-fold cross-validation.

Panel (a) of the figure is a schematic of exactly these five stages, so a reader can orient themselves before looking at the more detailed panels that follow.

4. Panel-by-Panel Walkthrough

Panel (a) Pipeline schematic

A simple flow diagram of the five stages described above (raw spectra → synthetic augmentation → PCA reduction → quantum angle encoding → hybrid classification). It doesn't contain data — it's purely an orientation aid so the rest of the figure is easier to follow.

Panel (b) Real vs. synthetic Raman spectra

This panel overlays real spectra (solid lines) against synthetically generated spectra (dashed lines) for three representative classes (M50, M32/M52, M54). The point of this panel is to visually justify the augmentation step if the synthetic spectra didn't resemble the real ones, expanding 70 samples to 14,000 would just be adding noise rather than genuinely useful training data. Because the dashed and solid traces track each other closely across the whole spectral range, this supports the claim that the statistical generation model preserves each class's true peak structure.

Panel (c) PCA explained variance

This bar/line chart shows how much of the total spectral variance each of the 8 retained principal components captures individually (green bars) and cumulatively (red line). It answers the question “why 8 components?” — the first two components alone already capture roughly 65% of the variance, and by the 8th component the cumulative curve has largely flattened, indicating diminishing returns from keeping more components. This is the empirical justification for fixing the qubit-ready representation at 8 dimensions rather than choosing that number arbitrarily.

Panel (d) Quantum-encoded feature space

Because the qubit-ready representation is 8-dimensional, it can't be plotted directly. This panel shows a 3D scatter of just 3 of the 8 rotation angles (θ_1 , θ_2 , θ_3), colored by glucose class. Even in this reduced 3-of-8 view, points from the same class tend to cluster together, which suggests the PCA-plus-angle-encoding step is preserving class-discriminative structure rather than destroying it during compression — consistent with the strong classification results that follow in panels (e) and (f).

Panel (e) Model performance comparison

A grouped bar chart of Accuracy, F1-score, and AUC for all nine classifiers under 5-fold cross-validation, giving a quick visual ranking that mirrors the numbers reported in Table 1. The ensemble methods (Voting, Stacking) and CatBoost sit at the top; AdaBoost is visibly the weakest. This panel is meant to convey the overall ranking pattern at a glance — the exact mean $\pm$ std values for the paper always come from Table 1, not from reading bar heights off this chart.

Panel (f) Micro-averaged ROC curves

Standard ROC curves for all nine classifiers, micro-averaged across the 14 glucose classes and across the 5 folds. Curves that hug the top-left corner (Voting Ensemble, CatBoost, Stacking Ensemble, Random Forest) indicate excellent class separation; Decision Tree and AdaBoost sit further from that corner, in agreement with their lower AUC scores in Table 1. This is the same information as the original single ROC figure, just placed alongside the rest of the pipeline for context.

5. Key Numbers Behind the Figure

For quick reference when discussing the figure, these are the headline results from Table 1 (5-fold CV, mean $\pm$ std):

Model	Accuracy	AUC-score
Stacking Ensemble (proposed)	0.9143 $\pm$ 0.0168	0.9968 $\pm$ 0.0011
Voting Ensemble (proposed)	0.9129 $\pm$ 0.0153	0.9974 $\pm$ 0.0008
CatBoost	0.9029 $\pm$ 0.0145	0.9974 $\pm$ 0.0008
Random Forest	0.9093 $\pm$ 0.0147	0.9969 $\pm$ 0.0008
K-Nearest Neighbors	0.8914 $\pm$ 0.0208	0.9826 $\pm$ 0.0048
Decision Tree	0.8479 $\pm$ 0.0207	0.9181 $\pm$ 0.0111
Support Vector Machine	0.8243 $\pm$ 0.0118	0.9914 $\pm$ 0.0011
Logistic Regression	0.7914 $\pm$ 0.0114	0.9853 $\pm$ 0.0011
AdaBoost	0.3736 $\pm$ 0.0955	0.8815 $\pm$ 0.0223

6. An Important Caveat to Discuss

The word “quantum” in the framework's name refers to how the features are prepared, not to any quantum computation that is actually executed. Concretely: PCA reduces the spectra to 8 components, and those components are rescaled into $[0, \pi]$ so they could be used as rotation angles for quantum gates (R_x , R_y , R_z). But in the current pipeline, no quantum circuit or quantum simulator is ever run — the angle-encoded values are simply handed to classical classifiers (Random Forest, CatBoost, etc.). This is an accepted and standard preprocessing strategy in the quantum machine learning literature (referred to as “qubit-ready” or “quantum-inspired” encoding), and the paper is careful to describe it this way and to list it explicitly as a limitation in the Conclusion — but it's worth flagging clearly here so there's no ambiguity in review or in conversation with the supervisor about what was and wasn't run on an actual quantum system.

7. Questions for Discussion

- Whether panel (d)'s class labels should be relabeled with the real class names (M50, M52, ...) instead of generic “Class 1...Class 14”, to stay consistent with panel (b) and the rest of the paper.
- Whether the supervisor is comfortable with the current framing of “quantum-inspired” given that no circuit is actually executed, or would prefer stronger caveats earlier in the paper (e.g., in the Introduction) rather than only in the Conclusion.

- Whether it's worth adding a future-work note about validating panel (d)'s class separation using all 8 encoded dimensions (e.g., via t-SNE/UMAP) rather than just 3 of the 8 raw angles.